

Development of an artificial intelligence (AI) marine scientific research model: The case of marine science and technology policy and its legitimate practice in Taiwan

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ABSTRACT

This study explores the intersection of artificial intelligence (AI), marine science, and legal governance in Taiwan, with the project Development and Application of Marine Exploration and Ecological Survey Technologies under Climate Change as a case study. It highlights Taiwan's integration of AI into biodiversity monitoring, underwater exploration, and environmental analysis, while addressing legal challenges in intellectual property, data protection, and national security. A governance model was developed to enhance the legality, security, and credibility of marine datasets through data collection policies, legitimating measures, and inspection protocols. Comparative insights from the EU, United States, Japan, and Korea stress the need for Taiwan to refine AI policies, strengthen cross-agency coordination, and align with international frameworks, particularly the UN Sustainable Development Goal 14 (SDG 14) and the Ocean Decade. Taiwan's experience provides a replicable prototype for emerging maritime AI nations, linking innovation with legal compliance and sustainable ocean governance.

1. Introduction

Technological advances have revolutionized our lives. From facial recognition systems in smartphones to self-driving cars, AI is frequently discussed. However, technological advances have grown in lockstep with environmental destruction. For example, human activities, such as emissions from thermal power plants and vehicles, are the major causes of global warming, which has spurred efforts to combat climate change. Since oceans serve as natural carbon sinks, maintaining steady biodiversity is one solution to the global climate crisis. Large-scale marine environmental protection enables marine creatures and plants to recuperate, facilitating CO₂ absorption.

Building on this foundation, it is necessary to situate Taiwan's policy discourse in the broader scholarly context. A keyword search on the *Marine Policy* website using the terms "marine scientific research Taiwan" and "marine science technology Taiwan" reveals that

Taiwanese scholars have contributed to policy-oriented research on marine science, focusing on the Ocean Basic Act, marine spatial planning, and technology development strategies. Nevertheless, the volume of Taiwan-related publications in leading international journals remains limited, suggesting that stronger engagement with comparative literature is required. Specifically, integrating Taiwan's experience with insights from Japan, Korea, the European Union, and the United States will both demonstrate academic rigor and enhance the international significance of this research [1].

The proposed model explicitly supports the long-term objectives of the UN Ocean Decade by embedding sustainable practices and responsible data use into marine research protocols. Specifically, actionable targets include clear guidelines on data legitimacy, AI-driven environmental monitoring, and governance mechanisms aligned with international standards.

Although Taiwan is characterized by plentiful and diverse marine

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resources, recent research has shown that rapid changes in the marine environment have caused a decline in marine biodiversity. The alteration of ocean temperatures leads to behavioral changes in marine creatures. To reverse this negative trend, a prospective research project titled ‘Development and application of marine exploration and ecological survey technologies under climate change’ was commissioned [2]. Given AI’s ability to collect large amounts of data and analyze information [3], the team employs AI to assist their research work, aiming to develop and integrate various Taiwanese technologies to create a versatile underwater exploration system for marine exploration and environmental analysis. For instance, the project utilizes advanced AI-driven image recognition algorithms to analyze marine biodiversity, and employs big data analytics to process vast oceanographic datasets, enhancing environmental monitoring accuracy.

Technological advancement is accompanied by social and legal questions [4,5]. For instance, AI’s complexity could lead to social concerns about discrimination and misuse [6–8]. In recent years, many states have announced new ethics and guidelines [9,10] for AI to regulate AI activities. However, few studies have discussed AI policy and regulation, especially in the marine field. To address the legality issue, this study begins with a brief overview of a research project in Taiwan as a case study and then discusses the issues and challenges facing the application of AI. Furthermore, a Model for Assessing the Compatibility between AI Marine Science Research and the Legal System is proposed in Section 3. This section includes artificial intelligence research and development guidelines and the working principles of data and model governance. Section 4 examines the legality of AI technology usage within each sub-project team. The final section presents the conclusion. Marine AI applications uniquely intersect with spatial and legal dimensions, particularly due to operations within Exclusive Economic Zones (EEZs), sovereign jurisdiction, and sensitive government-collected marine data. Unlike typical digital content regulation, marine AI governance requires specialized legal frameworks addressing data sovereignty, international maritime law, and environmental compliance.

Recent policy studies in Taiwan have highlighted both progress and challenges in implementing sustainable marine science. For example, the National Academy of Marine Research has issued multi-year policy reports mapping out marine technology priorities, while National Taiwan Ocean University has launched interdisciplinary projects on AI and ecological survey technologies. Internationally, comparative cases include Japan’s integrated ocean policy, which coordinates legal, scientific, and industrial efforts; Korea’s smart aquaculture initiatives, which combine AI and IoT; and the EU’s Marine Strategy Framework Directive, which mandates ecosystem-based management supported by scientific data infrastructures such as EMODnet. These examples collectively demonstrate that Taiwan’s initiatives must be evaluated alongside global practices to establish their legitimacy and relevance [11].

2. The research project: the application of AI to marine scientific research project development

2.1. Overview

The research project in 2021 called ‘Development and Application of Marine Exploration and Ecological Survey Technologies Under Climate Change’ was commissioned in 2021. It aimed to develop marine scientific solutions integrating various domestic technologies with artificial intelligence (AI) to create a diversified underwater system for marine exploration and environmental analysis. Each sub-project of the ‘Development and Application of Marine Exploration and Ecological Survey Technologies Under Climate Change’ is divided into many sub-projects, which are organized into three topics related to technological developments: topic A: ‘Ecological and Environmental Analysis System’, topic B: ‘Underwater Monitoring and Exploration System’, and topic C:

‘Underwater Vehicle Auxiliary System’. The number of sub-projects is ten, and each sub-project has its own names and research goals (A-1, A-2, A-3, B-1, B-2, B-3, C-1, C-2, C-3, and D).

Moreover, this project will also establish a shared platform, ‘Ocean Park’, to organize each dataset and AI model developed by the sub-projects [12]. Public users can directly upload images to utilize the AI technology developed by this project. When the project ends, it will influence technological application experiences and environmental conservation and promote legal advancements in AI technology development. To bridge the divide between the interdisciplinary discourses of the natural and social sciences, the title of sub-project D is ‘Research on the Social Impact and Legal Advancement of AI Applied to the Ocean’. Sub-project D aims to conduct research and analysis, provide suggestions on legal and policy aspects, reduce the societal impacts of AI applied to the ocean, propose suggestions on national marine policies, and provide amendment drafts of the Marine AI Act.

The legal checklist and governance model have been practically applied across several sub-project teams, receiving positive feedback. Teams reported enhanced clarity on IP and sensitive information compliance, although further legal assistance was sometimes necessary. This validation highlights the checklist’s practical utility and identifies areas for continuous refinement.

This paper will focus on the model established and the legality of AI technology for other research teams’ reference when using AI to conduct scientific research. Based on sub-project team (D) studies, the research scope pertains to each sub-project team’s research and team members to build a dataset that includes submarine geomorphology, water quality parameters, underwater sound, and information on marine creatures while addressing issues related to intellectual property rights and classified and sensitive information.

2.2. Issues and challenges facing the application of AI

The use of AI prompts invention and innovation, raising many novel intellectual property issues. AI’s creative thinking capacities can result in creating a patentable invention or infringing others’ patent rights [13]. For instance, text and data mining is a step in training machine learning systems or processes that can reproduce copyright-protected works without explicit authorization, resulting in infringement [14]. Thus, the application of AI leads to many challenges in the intellectual property field, such as how to address the problem of AI infringement on intellectual property rights and whether AI can count as an inventor or whether AI-generated works are protected by copyright.

Other issues related to AI include discrimination and fairness. Bias can be interpreted as ‘the action of supporting or opposing a particular person or thing unfairly, because of allowing personal opinions to influence your judgment’ [15]. As AI can conduct statistical analyses, the reasonableness of inferences made by AI largely depends on the database [16]. If a database reflects an underlying bias, the algorithm inevitably reflects it. In other words, AI programmers’ subjective value judgments may unknowingly be embedded in algorithms. In addition, specific trends in historical data could lead to AI to replicate inequalities or biases [17]. For example, some companies have started to use algorithms in recruitment decision-making, but AI could potentially favour particular job seekers based on personal information found on the Internet, which may not be fully accurate or representative of the applicants [18]. This could cause HR departments to miss out on many qualified employees. On the other hand, bias can also affect liberties. AI-driven systems have been used to help courts assess various risks, including parole decisions in some jurisdictions; however, intelligent software could exhibit racist or sexist tendencies if it learns racist or sexist information or extracts data from an unsuitable dataset [19].

The other issue discussed is the autonomy and control of AI. AI exerts only a supportive function in some tasks that are not mission-critical, such as data monitoring, and delivers the data to humans to make further decisions. AI can also be incorporated into mission-critical

Table 1
Blueprint of Marine Science and Technology Policy Development.

Goals		Intelligent Ocean		
Visions		Enhancing marine research capabilities, promoting sustainable ocean development		
Core Issues		Addressing global climate change		
Implement Strategies	1. Promoting marine industry development	1. Basic infrastructure development 1 A. Fostering marine information sharing and enhancing marine regulatory frameworks	2. Ensuring national security 1B. Encouraging marine technology innovation and advancing marine resource applications	3. Sustainable marine development 1 C. Deepening strategies for coastal and inland sea coexistence to foster industrial development environments
	2. Strengthening marine science and technology governance	2 A. Assessing gaps in marine science and technology, establishing research infrastructure	2B. Implementing marine science and technology enforcement, constructing national land protection systems	2 C. Utilizing emerging marine science and technology for diversified and harmonious development
	3. Implementing marine ecological and humanistic principles	3 A. Strengthening marine environmental surveys to ensure sustainable marine resources	3B. Achieving marine environmental protection, enhancing marine system resilience	3 C. Fostering nationwide marine education to enhance national marine awareness
The Application of Marine Technology		Advanced materials Sensor and imaging	Nanotechnology Satellite technology	Underwater engineering and technology Autonomous system
Infrastructure Resources		Establishment of marine basic data investigation research vessels / Ship model water tanks IoT / 5 G / B5G / 6 G infrastructure construction	Biotechnology Gig data analytics Comprehensive survey of Taiwan's entire marine area and cloud database	
		Establishment of regional marine key technology research centers		

systems, allowing it to make decisions. Thus, mechanisms that ensure effective human participation are important [20]. In particular, when AI causes harm or damage, a liability issue linked to algorithms emerges. Given that AI devices cannot be held civilly or criminally liable [21], apportioning liability among the manufacturer, programmer, and users of AI-enabled products has become a much-discussed topic [22].

In addition, the development of AI is associated with privacy and data governance. With the advent of AI facial recognition technology, surveillance activities have been augmented in some countries. This jeopardizes the privacy of people and harms society [16]. Furthermore, this kind of evolution also puts personal information in danger. Specifically, as the use of AI becomes more widespread, people's personal data can be gathered more easily, which prompts the need for data protection and governance frameworks, such as the European General Data Protection Regulation (GDPR), which aims to provide an appropriate framework for the control and protection of personal data, preventing the irresponsible use of AI that affects individual rights [23].

Transparency and traceability are crucial issues in AI governance. Considering that the data used by a computer directly affects the results under the AI applications, it also raises the question of how to ensure that the data and programming are accurate, especially since the large amount of data available on the Internet increases the difficulty of inspecting the programming that makes up the algorithm [21]. AI's capability to keep learning automatically without having to be reprogrammed by people can be divided into machine learning, deep learning, and neural networks, which focus on imitating human learning abilities using data and algorithms, to gradually improve their accuracy [24].

Given that an AI system can examine and detect additional patterns in large amounts of data to make better-automated decisions, it may have undergone numerous instances of data processing. This means that the answers generated by calculations could eventually outgrow the programmers' initial coding and become unknown to people [25]. As legal decision-making requires transparency to represent fairness to the public, this opacity can undermine legitimacy. People need to know the reasons for decisions, but the outcomes are generated without explanation under machine learning, and even programmers may not know how AI achieves its outcomes [16]. Therefore, various critics have raised concerns that the interpretability of decisions made using AI systems is essential [26].

3. A model for assessing the compatibility between AI marine science research and legal systems

3.1. Artificial intelligence research and development guidelines

To encourage the development of AI and breakthrough technology bottlenecks, the Ministry of Science and Technology (Taiwan) formulated AI Research and Development Guidelines [27], aiming to facilitate interactive dialogue between researchers and stakeholders, make research results more widely accepted, and create a reliable environment conforming to international trends in AI research.

The guidelines contain three core values: 'people-oriented,' 'sustainable development,' and 'diversity and inclusion,' which provide directions for Taiwan's AI researchers to follow. First, AI scientific research should always aim to improve human life, building an AI society that respects human dignity, freedom, and fundamental human rights. Second, AI scientific research should strive for a balance of interests among economic growth, social progress, and environmental protection, creating a sustainable vision to coexistence and co-prosperity among humans, society, and the environment. Third, AI scientific research should aim to foster and accommodate an AI society with diverse values and backgrounds while actively initiating a cross-disciplinary dialogue mechanism to benefit all people's understanding and cognition of AI.

As mentioned previously, despite AI being designed to help humans, its applications can harm society. These three goals facilitate a realistic understanding of where AI is likely to impact society. Additionally, the legal system faces challenges, as AI raises new complexities regarding employment, rights, duties, and accountability. Thus, organizations should develop regulations or reliable methods to govern inaccuracies, accidents, biases, and unexpected behaviors during the utilisation of AI, which can eventually mitigate the impact of AI development, enhance Taiwan's AI technological capabilities, and successfully use it in practical fields.

Comparative legal analysis suggests that Taiwan's guidelines align with global trends but remain relatively general. For example, the EU has incorporated AI ethics principles directly into legislative proposals such as the AI Act, while Japan's Cabinet Office has issued detailed AI R&D principles under the *Society 5.0* initiative. Taiwan's guidelines, by contrast, emphasize values without specifying enforcement or compliance mechanisms. Bridging this gap will require more precise institutional design, such as establishing mandatory compliance audits or linking funding eligibility to adherence with ethical principles [28].

Table 2

Taiwan's regulations regarding intellectual property rights that may involve the project team's research activities.

1. Copyright Act	
Article 3 (1)	Given that article 3 states: 'Work' means a creation that is within a literary, scientific, artistic, or other intellectual domain, each sub-project team needs to examine the data and materials collected by them, identifying whether these materials belong to literary, scientific, artistic or other academic creations.
Article 5	If sub-project teams use others' work, they should recognize the range of fields that these materials belong to.
Articles 6 and 7	The adaptation of an original work is a derivative work, and the selection and arrangement of materials is a compilation work, both of which are protected as independent works.
Article 10	The author of a work shall enjoy copyright upon completion of the work
Article 30	About the duration for the economic rights for the author.
Article 37	The economic rights holder may license others to exploit the work.
Articles 40–1	Joint economic rights in a work shall be exercised with the consent of all joint economic rights holders.
Article 42	Economic rights are extinguished upon expiration of the term of protection.
Articles 44–46	The scope of fair use of economic rights.
Article 80–2	Reasonable infringement of copyright.
Various economic rights and moral rights that may be involved in the team's collection of information	
Oral and literary works	The right of reproduction, public dictation, public broadcast, public performance, adaptation, edition, rental, public transmission, and distribution.
Photographic works	The right of reproduction, public broadcast, public display, adaptation, edition, rental, public transmission, distribution.
Pictorial and graphical works	The right of reproduction, public broadcast, adaptation, edition, rental, public transmission, distribution.
Audio-visual works:	The right of reproduction, public broadcast, public presentation, adaptation, edition, rental, public transmission, distribution.
Sound recordings:	The right of reproduction, public broadcast, claiming payment of remuneration for public performance, adaptation, edition, rental, public transmission, distribution.
2. Patent Act	
Article 2	The project team is required to check whether the data collected has been applied for an invention patent, utility model patent, or design patent.
Article 58(1)	The patentee of an invention patent has an exclusive right to prevent others from exploiting the invention without the patentee's consent.
Articles 59 and 60	Limitations of patent rights
(1) Patent Act related to patent licensing	
Article 62	The assigning, entrusting, licensing, or establishing of a pledge on a patent right by the patentee shall have no locus standi against any third party unless it is recorded with the specific patent agency.
Article 63	An exclusive licensee may sub-license a third party to exploit the licensed patent.
Article 64	Where a patent right is jointly owned, except for exploitation by each of the joint owners, it shall not be assigned, entrusted, licensed, pledged, or abandoned without the consent of all the joint owner(s).
Article 70	Extinguishment of patent right.
Article 87	Grounds for compulsory licensing.
3. Trade Secrets Act	
Article 2	Each sub-project team needs to check that the method, technique, process, formula, program, design, or other information used in the course of production, sales, or operations meet the following requirements: 1. It is not known to persons generally involved in the information of this type; 2. It has economic value, actual or potential, due to its secretive nature; 3. Its owner has taken reasonable measures to maintain its secrecy.
Article 7	An owner of a trade secret may grant a license to another for the use of the trade secret.
Article 10	A situation of misappropriating of a trade secret.

3.2. The working principles of data and model governance

The working principles of data and model governance established by sub-project (D) play an important role in enhancing the security, legality, and credibility of each sub-project team's dataset by using data collection policies, legitimizing measures, material inspection, and identifying relative legal norms identification. Their scope focuses on intellectual property rights and the classification of sensitive information related to each sub-project team's research outcomes. These working principles organize the content of regulations concerning the issues mentioned above and use a questionnaire survey to establish a connection between the field of natural and social science (Table 1).

3.2.1. Data collection policy regarding intellectual property protection

Ocean data science initiatives (ODSIs) are now critical for improving ocean governance by mobilizing data [29]. The working principles of data and model governance frame a policy that identifies the specific legal norms involved in the establishment of a dataset. After consulting with the project team members, sub-project team (D) realized that all of the sub-projects were highly involved in intellectual property rights, as well as classified and sensitive information. The former is associated with patent rights, copyrights, and trade secrets, especially in the utilization and design of AI tools; the latter is associated with nationally classified information and sensitive technology, particularly when the team conducts scientific research activities in the water where the government exercises sovereign jurisdiction.

The field of intellectual property rights includes the following regulations in Taiwan:

- Patent Act;
- Copyright Act;
- Trade Secrets Act.

The field of classified and sensitive information includes the following regulations of Taiwan:

- The Classified National Security Information Protection Act;
- Government-funded science and technology research projects of the National Core Security Control Operations Manual;
- National Security Act;
- Guidelines for the Confidentiality of scientific and technological materials protection.

To check if the data collected and the database are regulated by these regulations, each sub-project team must conduct a legal inspection of the research material before collecting, processing, and using it. The crucial concepts and articles related to Taiwan's regulations in the field of intellectual property rights will be analyzed and organized by the sub-project team (D) and further delivered to each sub-project team. These concepts and articles are introduced in Table 2 and Fig. 1. The crucial concepts and articles related to Taiwan's regulations in the field of classified and sensitive information will be analyzed and organized by sub-project team (D) and further delivered to each sub-project team. This is addressed in Table 3 and Fig. 2.

3.2.2. Legitimizing measures for intellectual property rights and classified and sensitive information protection

To enhance the security, legality, and credibility of each subproject team's datasets, it is necessary to review the data and materials to ensure that the legitimizing measures are implemented before the dataset is published. For example, if the data collected and used by the sub-project teams concern copyright, the team should lawfully obtain a license from the copyright owner or use the work within a reasonable scope. When others' patents are involved, the license should be obtained from the patent holder, and the assignment of rights for a patent application (or patent right) shall become effective upon mutual agreement of the

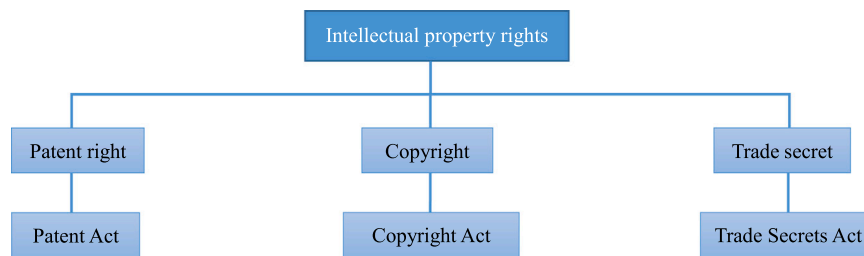


Fig. 1. Data collection policy regarding intellectual property rights.

parties concerned. However, the patent application or right will not be effective against any third party unless recorded at the Taiwan Intellectual Property Office (TIPO). If others' trade secrets are involved, authorization shall be obtained following the provisions of the Trade Secrets Act.

Concerning classified and sensitive information, if the information involves classified national security information or sensitive technology, it should be approved by the competent authority following the laws. The crucial concepts and articles of regulations for implementing legitimizing measures in intellectual property rights protection that involve sub-project teams are in Table 4 and Fig. 3. Crucial concepts and articles of regulations that involve implementing legitimizing measures in the field of classified and sensitive information protection for sub-project teams are in Table 5 and Fig. 4.

3.2.3. Materials inspection and relative legal norms identification of each sub-project team

When conducting research activities, sub-project teams should follow the Working Principles of Data and Model Governance established by sub-project team (D). Specifically, when sub-project teams understand the content of the Data Collection Policy, aside from executing the legitimizing measures, they should confirm whether intellectual property rights and classified and sensitive information are involved in data sources and fill out the material inspection form (Fig. 5).

Additionally, if other relevant legal bases and norms provided by sub-project team (D) apply to research activities during data collection, processing, and utilization, these regulations should be recorded and filled in the relevant legal norm identification form. The sub-project teams can consult legal professional members of the project team or the legal personnel of the project team's institution about any questions arising from data inspection.

Each sub-project team integrates research data and technologies to collect, process, and utilise underwater data to build a dataset that can significantly facilitate the application of AI in the marine field. Given that the research progress is different for each sub-project, the sub-project team (D) compiled the datasets provided by each sub-project team before 30 July 2020 to assist them in fulfilling the form for implementing risk management of data.

The form records all regulations that entail the research activities, comprehensively integrating them into the materials inspection form through a questionnaire form. If other sub-project teams submit datasets in the future, they will be supplemented sequentially (Table A1 and A2). This can help ensure that each dataset meets the regulations related to intellectual property rights, classified information, and sensitive technology, and thus achieve the purpose of establishing datasets that are secure, legal, and credible.

This procedural emphasis reflects global best practices in marine data governance. For instance, Canada's *Ocean Supercluster* requires formal compliance reports before data integration, while the EU's EMODnet mandates conformity with open-data directives and GDPR. Taiwan's inspection mechanism resembles these systems but must still evolve toward clearer enforcement structures, such as penalties for non-compliance and incentives for legal conformity [30].

4. The legality of AI technology usage of each sub-project

Vadrot et al. proposed drawing attention to the context of research practices in 'reflexive ocean science' [31]. Scientific laboratories, vessels, marine stations, various ocean-related data practices, and AI technology usage in the marine field need to comply with legal requirements.

First, studies of other sub-project teams indicate that the lawfulness of data must be ensured when developing or using AI. They must collect data during the research process, so a data collection policy for project members to follow helps satisfy this need. The formulation of data collection policies with the content of research projects mainly engages with the issue of intellectual property rights, including patent rights, copyrights, and trade secrets when using AI tools.

At the same time, conducting research in the marine area may also geographically involve classified national security information and sensitive technology. Consequently, the range of legitimate policies concentrates on these two fields.

Studies by the sub-project team (D) emphasize that marine scientific research must pay attention to the legality of AI technology usage. The sub-project team (D) collects and organizes the regulations' partial articles associated with specific legal action requirements when legitimizing measures, such as obtaining a license or authorization from the rights owner or competent authority.

This process forms a model that significantly facilitates interactive dialogue between researchers and stakeholders, making the research results widely accepted by society. Furthermore, this model focuses on intellectual property rights, privacy, data governance, and the country's sensitive information.

In Taiwan, the domain of intellectual property rights regulations encompasses the Patent Act, Copyright Act, and Trade Secrets Act. The regulation of sensitive information in Taiwan is categorized under National Security Information and Sensitive Technology.

This demonstrates that marine science must comply with the relevant authorities stipulated by legality. A historic opportunity now exists for Taiwan and other countries to reexamine the role of the rule of law in achieving a truly integrated, holistic, and transdisciplinary global ocean governance. Similar initiatives are visible in the EU's Marine Strategy Framework Directive and Japan's integrated ocean policy framework, both of which explicitly connect marine science to legal compliance. Taiwan's experience may thus serve as a regional test case with broader implications for international governance reform [32].

The legality of marine AI technology involves actions and measures ensuring that AI applied in marine technologies complies with legal regulations in their application and development, thereby protecting intellectual property rights, marine ecosystems, and national interests.

The research project team established a model that emphasizes the importance of the legality of marine technologies, especially in the application of AI within marine science. This model ensures legal compliance in intellectual property, privacy, data governance, and national security-sensitive information, which is essential for the ethical and lawful development of AI-driven marine technologies.

By following Taiwan's relevant legal frameworks, such as the Patent

Table 3
Taiwan’s regulations in the field of classified and sensitive information that may be involved in the project team’s research activities.

1. Classified National Security Information Protection Act	
Article 2	‘Classified national security information’ means that the information is owned by, or under the control of, the Government and requires protection against unauthorized disclosure. This designation occurs because of the classification level and to safeguard national security or national interest.
Article 4	Information may be classified in one of the three designations listed below: (1) Top Secret — shall be applied only to information, the unauthorized disclosure of which reasonably could be expected to cause exceptionally grave damage to the national security. (2) Secret — shall be applied only to information, the unauthorized disclosure of which reasonably could be expected to cause serious damage to the national security. (3) Confidential — shall be applied to information, the unauthorized disclosure of which reasonably could be expected to cause identifiable damage to the national security.
Article 6	When an employee of an agency that does not have original classification authority originates information believed to require classification, the information shall be protected in the manner prescribed by this Act and the Enforcement Rules. The information shall be transmitted promptly under appropriate safeguards to the agency which has appropriate subject matter interest and classification authority.
Articles 14–20	The control and safeguarding of classified national security information
Article 21	An agency shall not use classified information without obtaining an approval from the authority that originally made the approval.
Articles 32–35	Relative rules regarding the liability of keeping information confidential.
2. Government-funded science and technology research projects of the national core security controls operations manual (The materials protection of sensitive technology)	
Section 1	If the high-tech research results or data are divulged to other countries, mainland China, Hong Kong, and Macau, it could undermine the competitiveness of Taiwan; therefore, these materials should be considered as sensitive technology.
Section 1	The knowledge, theories, formulas, models, data, experimental records, process, technology, design drawings, samples, equipment, etc., which meet the above criteria and are created or value-added by the participants of each project team during the implementation of the project shall be kept confidential
3. Government-funded science and technology research projects of the national core security controls operations manual (The sensitive technology in marine field)	
Section 4	Underwater research: Environmental parameter of underwater acoustic research located in restricted and prohibited waters of Taiwan.
Section 4	Marine geology: Using multi-beam sonar to collect original water depth data which have not experienced the correction of attitude, tide level, and sound speed in Taiwan’s territorial waters, as well as the digital grid bathymetric data with a resolution of less than 200 m after correction processing. As for the outlying islands of Taiwan, the research areas are within restricted and prohibited waters.
Section 4	Ocean physics: Original hydrographic data in restricted and prohibited waters
4. National Security Act (sensitive technical data protection)	
Article 3 (1)	No person may engage in disclosing, delivering or transmitting confidential documents, drawings, images, messages, articles or electromagnetic records that are for official use for a foreign country, Mainland China, Hong Kong, Macau, foreign hostile forces, or various organizations, institutions, or groups established or substantially controlled by these organizations, institutions, or groups, or persons dispatched by them.
5. Guidelines for the confidentiality of scientific and technological materials protection	
Article 4	Confidentiality level of scientific and technological data.
Article 6	When government agencies at all levels refer to confidential information related to business needs, or when academic institutions, experts, scholars, manufacturers, and associations apply for access to confidential information, they shall obtain approval from the executive in charge according to the level of confidentiality.
Article 7	Measures to be taken by the confidentiality unit in the case of leakage or loss of scientific and technological data

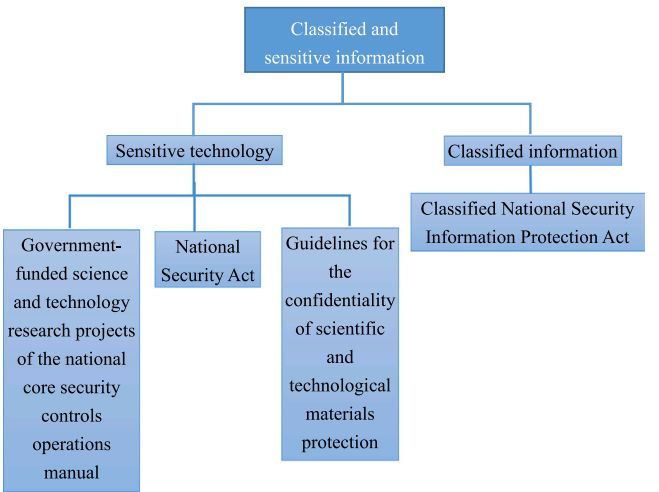


Fig. 2. Data collection policy regarding classified and sensitive information protection.

Act and National Security Act, the project seeks to foster a secure, legally compliant environment for AI research in marine sciences.

This approach not only facilitates interdisciplinary collaboration but also sets a precedent for legality-focused AI applications in global marine research.

5. Discussion

In 2019, Taiwan’s Ministry of Science and Technology (which became the Taiwan National Science and Technology Council, NSTC, in 2022) established the “AI Technology R&D Guidelines” [33]. In 2024, the Taiwan National Science and Technology Council announced the draft “Artificial Intelligence Basic Act” [34] in an attempt to keep up with the global legislative trend of regulating AI risks and safeguarding basic human rights. However, the current draft lacks a clear governance framework for AI research, development, and deployment. Notably, Article 5 stipulates that the “Interpretation and Application of Laws and Regulations” should follow the principle of “not hindering new technologies.” If enacted, this provision would elevate the Act above existing legal frameworks such as the Personal Data Protection Act, the Organic Act Governing the Establishment of the Financial Supervisory Commission, and the United Nations Convention on the Law of the Sea. This could significantly impact the stability and applicability of the existing legal system. Therefore, it is recommended that governments systematically review relevant domestic regulations, clarify the relationships between different laws, and identify areas requiring adjustments. This approach would enable an effective response to the challenges posed by AI, which is the core objective of this study and its advocacy.

On June 8, 2020, during National Ocean Day in Taiwan [35], the government released the National Ocean Policy White Paper [36] within one year of the enactment of the Ocean Basic Act [37]. The White Paper emphasized the construction of an AI-powered surveillance and intelligence network to detect and pre-empt potential threats, aiming to establish a comprehensive maritime defence system. It also promoted the integration of artificial intelligence, blockchain, cloud technology, big data analytics, ecosystem management, and the Internet of Things (IoT) into the marine industry to enhance smart operations and reduce costs. Furthermore, it highlighted the implementation of deep learning AI to strengthen computational capabilities, reduce manual labor, and improve research quality. Nearly three years after its release, the effectiveness of these AI-driven policies remains a critical issue. Given the rapid advancements in AI innovation and applications over the past three years, it is essential to assess whether the National Ocean Policy White Paper should be reviewed and revised to keep pace with these

Table 4
Regulations of legitimating measures in the field of intellectual property right protection that may involve the project team’s research activities.

1. Copyright Act (authorization from economic rights holder)	
Article 37	The economic rights holder may license others to exploit the work. The territory, term, content, method of exploitation, and other particulars of the license shall be as stipulated by the parties; particulars not clearly covered by such stipulations shall be presumed to have not been licensed.
2. Copyright Act (when sub-project teams are using material that may constitute a fair use on economic rights in the work)	
Article 44	The reasonable use of copyright for the purpose of legislation or administration by central or local government agencies.
Articles 48–1	Central or local government agencies, educational agencies that have been established by law, or libraries open to the public may reproduce abstracts appended to the following works where such works have been publicly released: 1. Masters theses or doctoral dissertations written under the Degree Conferral Act, where the author has obtained a degree. 2. Academic papers published in periodicals. 3. Research reports or collections of seminar papers that have been publicly released.
Article 50	Works publicly released in the name of a central or local government agency or public juristic person may, within a reasonable scope, be reproduced, publicly broadcast, or publicly transmitted.
Article 52	Within a reasonable scope, works that have been publicly released may be quoted where necessary for reports, comment, teaching, research, or other legitimate purposes.
Article 63	A situation that allows translating, adapting, and distributing the work of another person
Article 64	Lawfully exploiting the work of another person should clear indicate the source
Article 65	Fair use of a work shall not constitute an infringement on economic rights in the work. However, in determining whether the exploitation of a work complies with the reasonable scope in the Patent Act or other conditions of fair use, all circumstances shall be taken into account, and in particular, the following facts shall be noted as the basis for determination: (1) The purposes and nature of the exploitation, including whether such exploitation is of a commercial nature or is for non-profit educational purposes. (2) The nature of the work. (3) The amount and substantiality of the portion exploited in relation to the work as a whole. (4) Effect of the exploitation on the work’s current and potential market value.
5. Patent Act (Patent Rights)	
Article 62 (1)	The assigning, entrusting, licensing, or establishing of a pledge on a patent right by the patentee shall have no locus standi against any third party unless it is recorded with the specific patent agency.
Article 63	An exclusive licensee may sub-license a third party to exploit the licensed patent unless otherwise agreed upon by contract. A non-exclusive licensee shall not be allowed to sub-license a third party to exploit the licensed patent without the consent of the invention patentee or the exclusive licensee.
6. Trade Secrets Act	
Article 7 (1)	An owner of a trade secret may grant a license to another for the use of the trade secret. The territory, term, contents, methods of use or other matters in connection with the license shall be determined by the contract between the parties.

developments.

International comparisons illustrate the importance of continuous policy review. For example, the European Union has consistently updated its Marine Strategy Framework Directive and Digital Agenda to integrate AI into marine science while ensuring compliance with GDPR. Similarly, the United States’ National Oceanic and Atmospheric Administration (NOAA) AI Strategy emphasizes transparency and marine data sharing. Japan has advanced its *Society 5.0* vision, embedding AI into both industrial and environmental strategies, while Korea has launched AI-driven “smart ocean” programs focusing on aquaculture and real-time pollution monitoring. By contrast, Taiwan’s current White Paper lacks a detailed evaluation mechanism to assess the effectiveness of AI deployment in marine science and policy. This comparative gap highlights the urgency of revisiting Taiwan’s strategic documents to

remain aligned with international best practices [38].

The EU’s EMODnet provides a strong precedent by using AI for seabed classification under strict GDPR compliance, while the US NOAA AI Strategy emphasizes legal oversight and marine data sharing. These international frameworks offer valuable comparative insights, demonstrating the significance and originality of Taiwan’s marine AI governance model.

The author and the research team conducted the research project "Promotion and Implementation of Marine Technology Policy Guidelines and Action Plan" [39] in 2022 under the commission of the National Academy of Marine Research (NAMR), Taiwan [40]. This project systematically reviewed 27 key policy reports from 11 maritime-related administrative agencies and, under the framework of 9 major promotion strategies and 33 action plans, innovatively formulated 143 marine technology policy action plans. These plans serve as a reference for Taiwan’s short-, medium-, and long-term marine technology policy development and governance, including the following AI-driven marine technology policies.

5.1. Establishing homeland security protection [41] and marine monitoring system [42]

Developing Marine Intelligence System [43], strengthening advanced Optical Satellites [44] and Radar Satellites [45], installing Radar and Communication Systems [46], and developing Sensing Technology [47]. Through the integration of ships and Autonomous Underwater Vehicles (AUVs) [48], and utilizing Artificial Intelligence (AI), Internet of Things (IoT), and Marine Big Data Technologies, establish a real-time monitoring and response mechanism for marine conditions. This will help construct a Marine Information Security Early-Warning System [49] for defense, disaster response, administrative management, and marine development. Apply emerging marine technologies to assist in Technology Enforcement [50] in marine law enforcement.

Comparative evidence from Japan’s maritime surveillance architecture and Korea’s integrated radar-satellite systems underscores that Taiwan’s proposed mechanisms are feasible but require more explicit legal frameworks for inter-agency cooperation. In Japan, the Basic Act on Ocean Policy clearly mandates coordination between scientific research institutions and defense agencies, while in Korea, the Coast Guard has established joint data-sharing protocols with universities and private firms. Taiwan’s current approach lacks such formalized coordination, which may limit the operational effectiveness of its AI-driven monitoring systems [51].

5.2. Applying new maritime technologies [52] to assist in marine technology enforcement

To effectively implement marine safety, military operations, and humanitarian [53] rescue activities, enhance marine patrol capabilities and law enforcement capabilities, coordinate the protection and expansion of national marine rights and interests, suppress maritime violations (Maritime Torts [54]), safeguard Freedom of Navigation and Safety of Navigation in maritime areas, and consider the development and application of Advanced Materials [55], Autonomous Systems [56], Big Data Analytics [57], Human-Machine Interaction [58], Advanced Manufacturing [59], Energy Management [60], Electronic Warfare in Cyberspace [61], and human body functions.

The inclusion of cutting-edge technologies also calls for legal preparedness. For instance, the EU has already debated the legality of autonomous maritime vessels under the International Maritime Organization (IMO), while the US Navy actively experiments with unmanned surface vehicles regulated under strict liability standards. Taiwan’s strategy papers acknowledge emerging technologies but do not yet provide clear liability rules for accidents involving autonomous systems. Bridging this legal gap would increase both operational legitimacy and

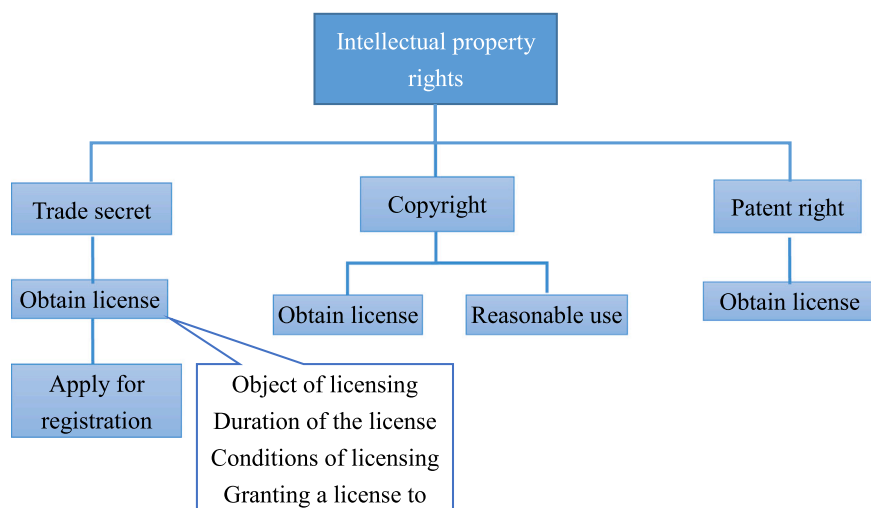


Fig. 3. The legitimating measure for intellectual property right protection.

international credibility [62].

5.3. Research and development of marine smart management system [63], ensuring harbor and vessel navigation and operational safety

Combining industry, government, academic, and business energy to apply Big Data and IoT technologies to vessel operations and ports, promoting Intelligent Navigation Safety System [64], such as Automatic Identification System (AIS) [65], Fishing Vessel Monitoring System (VSM) [66], and Long-Range Identification and Tracking System (LRIT) [67] devices and functions. This will improve navigation safety monitoring and maritime early warning capabilities for emergency response. At the same time, integrate the operating capacity of our national shipping companies and existing industry advantages, absorb international advanced technologies, and build upon domestic R&D achievements to develop new technologies and methods for Marine Environmental Monitoring.

Other jurisdictions provide instructive parallels. The EU has integrated its AIS and LRIT frameworks with environmental monitoring programs, while Singapore has built smart ports with AI-enabled logistics and legal safeguards for data use. Taiwan's initiatives, by comparison, remain fragmented across agencies. A more coherent framework—anchored in clear rules for data governance, privacy, and liability—would ensure that Taiwan's smart port development aligns with global legal standards [68].

5.4. Expanding regional and global cooperation

In addition to national reforms, Taiwan must strengthen participation in regional and global governance frameworks. For instance, the Partnerships in Environmental Management for the Seas of East Asia (PEMSEA) has encouraged the integration of digital technologies in regional marine governance, and APEC has launched initiatives on AI-enabled sustainable ocean management. Although Taiwan participates in APEC, its contributions on AI and marine science remain limited. Increasing its visibility through policy papers, joint projects, and comparative studies would reinforce Taiwan's international role. Moreover, Canada and Australia provide valuable models: Canada's *Ocean Supercluster* invests in AI-driven marine innovation with strong industry-academic collaboration, while Australia's CSIRO develops AI-based monitoring under a clear legal framework that integrates environmental law with data governance. These cases highlight the importance of embedding Taiwan's research into global dialogues to ensure both policy relevance and diplomatic visibility [69].

5.5. Addressing domestic legal and institutional gaps

Finally, Taiwan's domestic system still faces key challenges. The Ocean Basic Act provides only broad policy principles and does not yet establish binding standards for AI in marine science. Similarly, Taiwan's Personal Data Protection Act lacks sector-specific provisions for marine datasets, leaving uncertainties regarding privacy and security. Cross-ministerial coordination remains weak, with overlapping jurisdictions between the Ocean Affairs Council, Ministry of Digital Affairs, and National Science and Technology Council. A future Marine AI Act could address these deficiencies by: (1) defining legal responsibilities for different actors in AI-driven marine science, (2) mandating ethical and legal training for research teams, and (3) creating a unified oversight body to coordinate marine AI governance. Without these reforms, Taiwan risks lagging behind its peers, undermining both policy effectiveness and international credibility [70].

6. Conclusion

It also must recognize that shared information and joint knowledge development facilitate rapid transformation in the social-ecological ocean system [71]. However, the negative effects created by new marine science and technology, such as AI, are comprehensive and occur in the ethical, societal, and legal systems. Thus, it is essential to take measures to manage the utilization of AI.

Given that the amount of data that needs to be processed is increasing, AI will soon become integrated into human society. The downside of technological development not only exists in traditional areas of technology, such as copyright and patent rights, but also creates concerns in the ethical and social realms, ranging from transparency to human rights. Therefore, it is essential to assess the influence of AI and establish a mechanism that can help create an AI society where people live harmoniously.

In this paper, we introduced a model that can assess the compatibility between AI marine science research and the legal system, hoping to support and encourage scientists to conduct research activities with the suitable use of AI. Taiwan's governance model can serve as a replicable 'policy prototype' for other emerging maritime AI nations, especially those with similar legal-system capacities and institutional constraints, facilitating the integration of interdisciplinary legal audit mechanisms within AI research initiatives.

The interdisciplinary dialogues between natural and social sciences are crucial during this rapidly developing high-tech era. In the context of the 'Development and application of marine exploration and ecological

Table 5
Regulations of legitimating measure in the field of classified and sensitive information protection that may involve in the project team’s research activities.

1. Classified Information	
Classified National Security Information Protection Act (The use of classified national security information should obtain approval beforehand.)	
Article 14	Other than persons who handle the information within his or her scope of official duty, classified information cannot only be known, possessed or used without a written authorization or approval from the original classification authority or its superior agency.
Article 21	An agency shall not use classified information without obtaining an approval from the original classification authority.
Classified National Security Information Protection Act	
Article 32	Liability concerning disclosing or delivering of classified national security information as approved by this law
Article 33	Liability concerning disclosing or delivering classified national security information petitioned for approval as classified national security information.
Article 34	Spying or collecting of classified national security information approved under this law.
Article 35	Liability of destroying, damaging, or concealing information properly classified under this Act, making such information unavailable. the liability of destroying, damaging, or concealing information properly classified under this Act, making such information unavailable.
National Security Act	
Article 7	The liability of any person who intends to endanger national security or social stability and commits the following offenses on the part of Mainland China: 1. Initiating, funding, hosting, manipulating, directing or developing an organization. 2. Disclosing, delivering or transmitting confidential documents, drawings, images, messages, articles or electromagnetic records that are for official use. 3. Spying on or collecting confidential documents, drawings, images, messages, articles or electromagnetic records that are for official use.
2. Sensitive technology	
Guidelines for the confidentiality of scientific and technological materials protection (Access and use of sensitive technical information must be approved in advance.)	
Article 6 (2)	Government agencies at all levels refer to confidential information due to business needs, or when academic institutions, experts, scholars, manufacturers, and associations apply for access to confidential information, they shall obtain approval from the executive in charge according to the level of confidentiality.
Government-funded science and technology research projects of the national core security controls operations manual (Sensitive technology control and confidentiality measures)	
Section 5	If the government-funded science and technology research project is identified as a sensitive technology research project, the project team should state this in its funding contract and the research results and materials should be controlled by competent authority.
Section 5	The project team should sign a non-disclosure agreement with the sensitive technology project researchers.
Guidelines for the confidentiality of scientific and technological materials protection	
Article 6 (3)	The exchange of confidential information with foreign government agencies, academic institutions, non-governmental organizations, experts, scholars, or general persons shall be carried out after obtaining approval from the executive in charge according to the level of confidentiality.
Article 6 (7)	When investing in foreign countries or conducting technical cooperation, data that are identified as ‘extremely confidential’ by the competent government ministry shall not be exported without permission of the competent ministry. For the export of technical data that is classified as confidential, the contract or agreement for technology export shall specify the confidentiality provisions in the process of technology transfer and require the technical personnel involved in the export or technical cooperation plan to be aware of the confidentiality provisions, or sign a confidentiality agreement that forbids technical personnel from disclosing technical information which exceeds the scope of the contract or agreement.
Article 6 (9)	When the staff of each unit publishes papers or participates in seminars at academic conferences, the report should be limited to basic theories and results and if technical details and data are involved, approval of the competent person must be acquired according to the confidentiality level of the information.
Article 6 (13)	Each unit shall request that new technical personnel sign a security agreement that forbids them to disclose secrets or collect and hide classified information after resignation. In the event of a breach,

Table 5 (continued)
offenders should be investigated and punished in accordance with relevant laws and regulations.

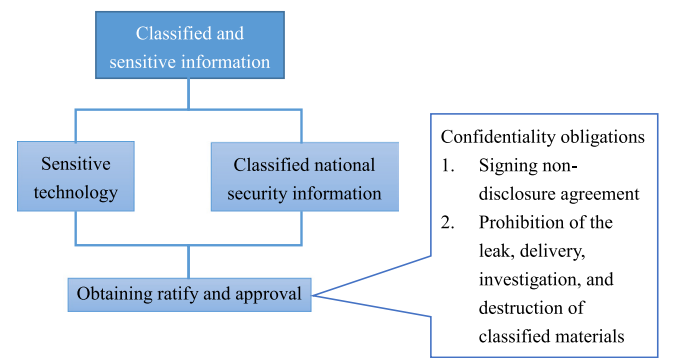


Fig. 4. The legitimating measure for classified and sensitive information protection.

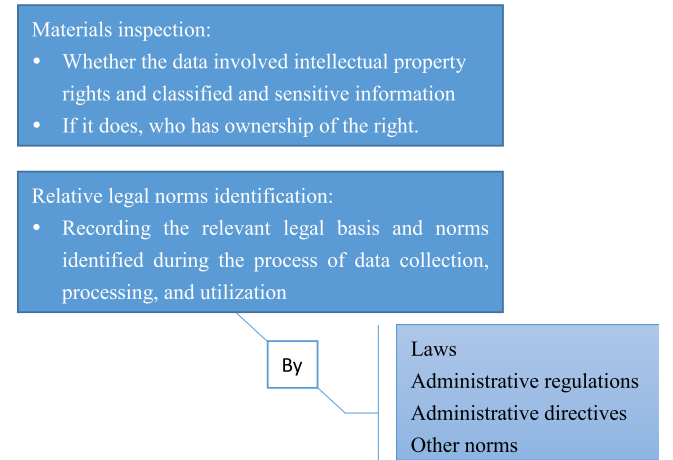


Fig. 5. Materials inspection and relative legal norms identification on intelligent property rights and classified and sensitive information.

survey technologies under climate change’ project commissioned by Taiwan’s Ministry of Science and Technology, the National Taiwan Ocean University’s team formed a legal working group. Sub-project (D) established the Working Principles of Data and Model Governance to achieve this goal, which consists of three parts: (a) data collection policy, (b) legitimating measures, and (c) material inspection and relative legal norm identification.

Apart from reconfirming that the regulations apply to the research activities, the material inspection and identification of relative legal norms also promote the security, legality, and credibility of each sub-team dataset to be published on the shared platform. This study forms a model that significantly facilitates interactive dialogue between researchers and stakeholders, making the research results widely accepted by society. It also bridges the gap between the two fields’ understanding, helping scientists comply with applicable laws and regulations. Additionally, it shows that the Ocean Decade and Marine Science and Technology Policy Development should prioritize the legality of AI technology usage when performing marine scientific research.

Concrete recommendations for policy makers include establishing formal regulatory guidelines for AI use in marine research, mandatory legal and ethical training for researchers, and creating standardized protocols aligned with international practices, particularly by the Ocean Affairs Council and NAMR. Future studies should also expand beyond

Taiwan by incorporating cross-country comparisons and empirical surveys with stakeholders, ensuring that the governance framework for AI in marine science remains adaptive, legitimate, and globally relevant [72].

This research has some limitations and implications for future studies in AI marine science. Firstly, the study is based on a single AI marine research project in Taiwan. Future research could expand the sample to include interviewees from other countries, enabling comparative analyses to better understand the legality of AI technology usage. Additionally, the application of AI technology in marine science varies due to differing national policies and regulations. Future research should consider examining developed countries with comprehensive policies and regulations applicable in this context.

Author statement

I, Wen-Hong Liu, certify that Prof. Juei-Cheng Jao and I have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript. Chiao-Wen Lin, Pin-Chien Chen, Yi-Cheng Chang and Yi-Juen Chen help collected and analyzed data and then conducted some sections. Furthermore, I certify that this manuscript has not been submitted to or published in any other publication.

CRedit authorship contribution statement

Yi-Cheng Chang: Writing – original draft, Visualization, Investigation, Data curation. **Wen-Hong Liu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Chiao-Wen Lin:** Writing – original draft, Visualization, Investigation, Data curation. **Juei-Cheng Jao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yi-Juen Chen:** Writing – review & editing, Writing – original draft, Validation, Conceptualization. **Pin-Chien Chen:** Writing – original draft, Visualization, Investigation, Data curation.

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Appendix A. Supporting information

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Data availability

The data that has been used is confidential.

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